

Sugumaran Balasubramaniyan

LLM & Agentic AI Systems Engineer | AI Platform & MLOps

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Summary

Senior data science and ML engineering professional with 7+ years of experience building AI applications, production data pipelines, and enterprise workflows in consulting, enterprise, and academic environments. Focused on LLM and agentic AI systems, ML governance, MLOps, and cloud infrastructure for mission-critical and regulated use cases.

Technical Skills

- **LLM & Agentic AI:** Python, LangChain, LangGraph, RAG, Hugging Face, Transformers, BERT, AWS Bedrock, LLM evaluation, agent orchestration.
- **ML & MLOps:** PyTorch, MLflow (tracking, registry), model monitoring and drift detection, AWS SageMaker, Databricks, CI/CD, REST APIs, data pipelines.
- **Cloud & Infra:** AWS (Glue, Lambda, S3, IAM), Kubernetes, Docker, Airflow, SQL/NoSQL, PySpark, ETL/ELT.
- **Governance:** Model governance, GDPR-ready data modeling, EU AI Act readiness, SHAP/LIME, audit trails.

Experience

Professor – Data Analytics & AI/ML

Jan 2026 – Present

SKEMA Business School, Paris, France

- Lead instructor for data analytics, AI, SQL, Python, and ML, emphasizing production-oriented and reproducible workflows.
- Designed projects around data pipelines, ML systems, and deployment-oriented analytics using cloud-based platforms.

AI / ML Engineer

Jan 2024 – Dec 2025

The Validate, Paris, France

- Architected AWS data pipelines (Glue, Lambda, S3) to process enterprise datasets in production.
- Implemented MLflow-based governance and GDPR-aligned tracking on Databricks and SageMaker for experiment traceability and model versioning.
- Built agentic AI workflows with LLMs, LangChain/LangGraph, Claude Code, and n8n for AI orchestration systems.

Data Analyst / Engineer

Jul 2023 – Dec 2023

Joubert Associés, Paris, France

- Designed Power BI dashboards tracking 50+ projects and optimized SQL, reducing refresh time by 36%.
- Led A/B testing for marketing campaigns, increasing lead conversion from 8% to 12%.

Data / AI Engineer

Apr 2021 – Jul 2022

Capgemini Technology Services, Chennai, India

- Developed REST APIs and ETL pipelines with AWS Glue and Python, improving AI/ML workflow speed by 45%.
- Implemented MLflow-based governance and drift monitoring, reducing model drift by 18% and establishing versioning with AWS SageMaker for Philip Morris International.

Data Analyst

Mar 2020 – Apr 2021

Infosys, Chennai, India

- Conducted risk management analysis for Citizens Bank (USA) using Python and SQL, identifying credit and operational risks and mentoring junior analysts.

Analyst

Jun 2018 – Feb 2020

Arimalytics, Puducherry, India

- Improved forecast accuracy by 12% for a manufacturing client through predictive inventory and maintenance models.

Education

MSc Data Science & AI Strategy, emlyon business school, Paris, France

Sep 2022 – Feb 2024

Exchange (Data Science, IoT, Big Data Analytics), McGill University, Montréal, Canada

May 2023 – Jul 2023

MBA, Pondicherry University, India

Jul 2016 – May 2018

Bachelor of Technology, Pondicherry University, India

Jul 2012 – May 2016

Certifications

AWS Machine Learning Engineer • Databricks Generative AI Engineer • NVIDIA Agentic AI (Professional) • NVIDIA Generative AI LLMs • Power BI (PL-300) • Dataiku Generative AI Practitioner • Dataiku ML/MLOps Practitioner • AWS Solutions Architect Professional (in progress) • Atlassian Agile Project Management Professional

Languages & Work Authorization

Languages: English (fluent), French (professional), Tamil (native). **Work authorization:** APS (France)